
1. The Bases are Covered

Program Name: Bases.java

Input File: bases.dat

Judges Input File

39

Judges Output to Screen

Base 10	Base 2	Base 4	Base 8	Base 12	Base 16
1	1	1	1	1	1
2	10	2	2	2	2
3	11	3	3	3	3
4	100	10	4	4	4
5	101	11	5	5	5
6	110	12	6	6	6
7	111	13	7	7	7
8	1000	20	10	8	8
9	1001	21	11	9	9
10	1010	22	12	a	a
11	1011	23	13	b	b
12	1100	30	14	10	c
13	1101	31	15	11	d
14	1110	32	16	12	e
15	1111	33	17	13	f
16	10000	100	20	14	10
17	10001	101	21	15	11
18	10010	102	22	16	12
19	10011	103	23	17	13
20	10100	110	24	18	14
21	10101	111	25	19	15
22	10110	112	26	1a	16
23	10111	113	27	1b	17
24	11000	120	30	20	18
25	11001	121	31	21	19
26	11010	122	32	22	1a
27	11011	123	33	23	1b
28	11100	130	34	24	1c
29	11101	131	35	25	1d
30	11110	132	36	26	1e
31	11111	133	37	27	1f
32	100000	200	40	28	20
33	100001	201	41	29	21
34	100010	202	42	2a	22
35	100011	203	43	2b	23
36	100100	210	44	30	24
37	100101	211	45	31	25
38	100110	212	46	32	26
39	100111	213	47	33	27

Note to Judges: The width of the columns is not important as long as the data in the columns are left justified and there is at least one space between any two numbers in a row.

2. Braille Student

Program Name: Braille.java

Input File: braille.dat

Judges Input File

5

```
1000111011001010010001100000111010100100
0000010111001100100001010000101001001011
0000001000000000100011000001001000001001
00111011111001101000000001110110111110
00000100100011011100000010001001010001
01001010101110001000010110000000100000
000010011000001010111001000111000110011010001000
000000101001001101000010001001000110110010010011
010111001001001010001010000010001100000000010001
01011100111000010011
11100100010100110000
11001000101001110011
00001101101001110010001010011010001000010111100010
00001111011111010010011100101010000100100101010000
01011011001010110000011000001010010000001100000100
```

Judges Output to Screen

```
a dog has 5 Fleas.
Computer SCIENCE
UIL rocks in 2012a.
win noW x
QWERTY Braille i45A
```

3. Chutes

Program Name: Chutes.java

Input File: chutes.dat

Judges Input File

```
4
2
3735928559
4 4
27 10
10 6
57 41
34 14
14 37
17 49
50 64
22 37
3
4206243583
3 4
61 45
60 45
45 37
17 48
4 54
11 39
40 56
4
3132799674
0 0
1
3073428133
11 7
3 1
11 3
44 11
63 18
62 51
61 50
60 17
43 10
55 27
46 20
59 14
27 42
36 57
32 33
49 63
18 33
33 48
9 12
```

Note to judges: below are the random numbers generated until the final roll for each seed in the input file

3735928559

2 3 5 6 4 2 6 4 4 4 5 4 2
4 3 5 2 4 4 2 5 1 4 5 5

4206243583

2 2 5 6 2 2 1 2 6 5 4 3 6
4 6 4 1 4 1 5 3 1 2 4 2 6
6 2 5 6 6 3

3132799674

4 4 5 3 5 6 3 5 3 6 5 1 5
5 3 5 6 5 6 2 6 3 4 6 5 3
6 5 1 5 1 3 2 2 4 4 5 3 3
3 2 1 5 2 1 5 3 2 4 2 2 2
4 6 5 5 5 6 4 3 4 4

3073428133

6 3 4 1 5 6 5 1 4 5 4 3 4
1 4 6 6 1 2 6 2 6

Judges Output to Screen

```
Player A wins after 25 rolls!
Player B wins after 32 rolls!
Player B wins after 62 rolls!
Player A wins after 22 rolls!
```

4. Emergency Room

Program Name: Emergency.java

Input File: emergency.dat

Judges Input File

```
4
8 5
00:01 9
00:01 9
00:01 3
00:06 8
11:01 3
11:15 8
11:30 8
23:10 1
18 10
00:01 9
00:01 9
00:01 3
00:06 8
01:01 3
01:15 8
02:30 8
03:00 1
03:01 9
03:01 9
03:15 3
03:20 4
05:01 3
05:06 8
11:01 3
11:15 8
11:30 8
23:10 1
8 5
00:01 3
00:01 3
00:01 3
00:01 3
00:01 3
00:01 3
00:01 3
00:01 3
00:01 3
9 5
00:00 1
00:00 1
00:00 1
00:13 1
00:13 1
00:13 1
00:26 1
00:26 1
00:26 1
```

Judges Output to Screen (cont. on next page)

4. Emergency Room (cont.)

Judges Output to Screen

4 doctors
4 doctors
8 doctors
3 doctors

Program Name: Etch.java Input File: etch.dat

Judges Input File

```

4
5 5
R10 D10 L5 U5 R10 D10 U5 L3 U5 R4 U13 L22 D5 R4
10 15
L12 D5 R5 D5 L5 D4 R15 U6 L5 R3 D8
1 1
U12 R3 D10 R3 U8 R3 D5 R3 U3 L10
7 6
U5 D8 R3 L6 D4 U5 D5 U6 R4 U4 L3 D2

```

Judges Output to Screen

[illegible]

(continued on next page)

5. Etch-a-Sketch (cont.)

Judges Output to Screen (cont.)

[illegible]

6. Message

Program Name: Message.java **Input File:** message.dat

Judges Input File

4

4

MERRY EASTER TO EVERY ONE NAME LICKETY SPLIT IS A HAPPY WORLD MIKE MAY HAVETO
CONVEY A LIVELY WATERING HOLES TO LIVE IN A PINKISH RIBBON OF CANDY. TOM ROSS
HAS A LOVELY, WHITE HOUSE ON THE END OF THE MEADOW ON LINDSEY LANE IN
BEAUMONT, TEXAS.

5

RICHARD CAN HAVE A VERY LONG PARIS TRIP TO THE END OF THE BORDER BUT BETTER
HOMES CAN BE FOUNDED ON THE WATER. THE NICEST BANKS IN THE LAND ARE NEAR THE
EDGE OF THE RED RIVERR. THE BUSHES ARE NEXTT TO THE WEDES IN THE MIDDLE OF
THE LAKE. I CAN SEEYA HAVE A LOT OF BUSINESS TO DO SO PLEASE ENJOYY
DECIPHERING THIS NOTE.

3

I AM COZY NOW THAT MY .COM COMPANY IS ANXIOUS TO BE INVESTIGATED TO SEE WHAT
IN THEE WORLD IS HAPPENING. I BELIEVE THAT JON KNOWS A BETTER BET THAN YOU
COULD EVER BELIEVE IN CNT OF WORLDLY EVENTS.

4

ROSS IS THE BETTER STREET TO FLOUNDER UP THE HILL. WHERE THERE ARE FOUR OR
FOUR OR MAYBE FIVE, YOU CAN CERTAINLY REASON HOW LIND CAN BE AVAILABLE TO
FIND THE FUNNIEST RIVER IN THE WORLD.

Note: Although the paragraphs in example input above appear on the printed page to consist of more than one line, each one is actually a single line in the input file.

Judges Output to Screen

NAME MIKE LIVE ROSS LANE
PARIS HOMES WATER BANKS NEXTT WEDES SEEYA
NOW COM SEE JON BET YOU CNT
ROSS HILL FOUR FOUR FIVE LIND FIND

Note to Judges: Extraneous spaces at the end each line of output is ok.

7. Most and Least

Program Name: MostLeast.java

Input File: mostleast.dat

Judges Input File

5

MARY HAD A LITTLE LAMB, LITTLE LAMB, LITTLE LAMB, MARY HAD A LITTLE LAMB, ITS FLEECE WAS WHITE AS SNOW. AND EVERYWHERE THAT MARY WENT, MARY WENT, MARY WENT, AND EVERYWHERE THAT MARY WENT, THE LAMB WAS SURE TO GO.
BAA, BAA, BLACK SHEEP, HAVE YOU ANY WOOL? YES SIR, YES SIR, THREE BAGS FULL.
ONE FOR THE MASTER, AND ONE FOR THE DAME, AND ONE FOR THE LITTLE BOY WHO LIVES DOWN THE LANE. ONE FOR THE MASTER, AND ONE FOR THE DAME, AND ONE FOR THE LITTLE BOY WHO LIVES DOWN THE LANE.
FIVE LITTLE DUCKS WENT OUT TO PLAY OVER THE HILL AND FAR AWAY THE MOTHER DUCK SAID QUACK, QUACK COME BACK FOUR LITTLE DUCKS CAME RUNNING BACK FOUR LITTLE DUCKS WENT OUT TO PLAY OVER THE HILL AND FAR AWAY THE MOTHER DUCK SAID QUACK, QUACK COME BACK THREE LITTLE DUCKS CAME RUNNING BACK THREE LITTLE DUCKS WENT OUT TO PLAY OVER THE HILL AND FAR AWAY THE MOTHER DUCK SAID QUACK, QUACK COME BACK TWO LITTLE DUCKS CAME RUNNING BACK TWO LITTLE DUCKS WENT OUT TO PLAY OVER THE HILL AND FAR AWAY THE MOTHER DUCK SAID QUACK, QUACK COME BACK NO LITTLE DUCKS CAME RUNNING BACK SAD MOTHER DUCK WENT OUT ONE DAY OVER THE HILL AND FAR AWAY THE DADDY DUCK YELLED QUACK, QUACK COME BACK. FIVE LITTLE DUCKS CAME RUNNING BACK.
LONDON BRIDGE IS FALLING DOWN, FALLING DOWN, FALLING DOWN. LONDON BRIDGE IS FALLING DOWN, MY FAIR LADY. TAKE A KEY AND LOCK HER UP, LOCK HER UP, LOCK HER UP. TAKE A KEY AND LOCK HER UP, MY FAIR LADY.
MY DOG HAS FLEAS. MY DOG HAS FLEAS. MY CAT.

Note: Although the paragraphs in example input above appear on the printed page to consist of more than one line, each one is actually a single line in the input file.

Judges Output to Screen

6 MARY
1 AS FLEECE GO ITS SNOW SURE THE TO WHITE
8 THE
1 ANY BAGS BLACK FULL HAVE SHEEP THREE WOOL YOU
10 BACK QUACK THE
1 DADDY DAY NO ONE SAD YELLED
4 DOWN FALLING HER LOCK UP
2 A AND BRIDGE FAIR IS KEY LADY LONDON MY TAKE
3 MY
1 CAT

8. Petite Pals

Program Name: PetitePals.java

Input File: petitepals.dat

Judges Input File

```
10
RADAR
STATEROOM
ATOYOTA
GOHANGASALAMIIMALASAGNAHOG
ABBACDDEFGGF
ABACDCFERABA
ABCDEDCBA
BBAADDEEFFGG
ROGER
AA
```

Judges Output to Screen

```
4
9
6
23
12
9
8
12
4
1
```

9. Rasterizer

Program Name: Rasterizer.java

Input File: rasterizer.dat

Judges Input File

```
4
10 8
BOX W 1 1 3 4 Y
BOX A 9 7 100 100 N
LINE L 2 4 8 4
CROSS X 5 4 4 4
END
16 9
BOX T 1 1 14 7 N
LINE C -1 -1 5 -1
LINE D -1 5 6 5
CROSS Z 7 4 3 3
BOX A 10 2 13 6 Y
END
1 1
END
4 4
BOX A 0 0 3 3 N
BOX B 1 1 2 2 N
BOX C 2 2 2 2 Y
LINE D 1 1 1 1
END
```

Judges Output to Screen

```
.....
.WWW.....
.WWW.....
.WWW.X....
.WLLXXXXL.
.....X....
.....X....
.....A

.....
.TTTTTTTTTTTTTT.
.T.....AAAAT.
.T.....Z..AAAAT.
.T....ZZZ.AAAAT.
DDDDDDZ..AAAAT.
.T.....AAAAT.
.TTTTTTTTTTTTTTT.
.....

.

AAAA
ADBA
ABCA
AAAA
```

10. Remnants

Program Name: Remnants.java

Input File: remnants.dat

Judges Input File

7 120 14 2 99 51 70 42 55

Judges Output to Screen

2
26
3
1
22
11
15
9
12

11. Spiral Galaxies

Program Name: Spiral.java

Input File: spiral.dat

Judges Input File

22 33 13 49 429 601 625 468 400 198 265 177 257 7

Judges Output to Screen

4 5
3 1
1 1
1 7
13 21
25 25
1 25
6 1
20 1
15 2
17 9
1 7
17 1
3 3

(continued on next page)

11. Spiral Galaxies (cont.)

Note to judges: this spiral galaxy can be used for x ($1 < x \leq 625$)

553	552	551	550	549	548	547	546	545	544	543	542	541	540	539	538	537	536	535	534	533	532	531	530	625
554	463	462	461	460	459	458	457	456	455	454	453	452	451	450	449	448	447	446	445	444	443	442	441	624
555	464	381	380	379	378	377	376	375	374	373	372	371	370	369	368	367	366	365	364	363	362	441	528	623
556	465	382	307	306	305	304	303	302	301	300	299	298	297	296	295	294	293	292	291	290	361	440	527	622
557	466	383	308	241	240	239	238	237	236	235	234	233	232	231	230	229	228	227	226	289	360	439	526	621
558	467	384	309	242	183	182	181	180	179	178	177	176	175	174	173	172	171	170	225	288	359	438	525	620
559	468	385	310	243	184	133	132	131	130	129	128	127	126	125	124	123	122	169	224	287	358	437	524	619
560	469	386	311	244	185	134	91	90	89	88	87	86	85	84	83	82	121	168	223	286	357	436	523	618
561	470	387	312	245	186	135	92	57	56	55	54	53	52	51	50	81	120	167	222	285	356	435	522	617
562	471	388	313	246	187	136	93	58	31	30	29	28	27	26	49	80	119	166	221	284	355	434	521	616
563	472	389	314	247	188	137	94	59	32	13	12	11	10	25	48	79	118	165	220	283	354	433	520	615
564	473	390	315	248	189	138	95	60	33	14	3	2	9	24	47	78	117	164	219	282	353	432	519	614
565	474	391	316	249	190	139	96	61	34	15	4	1	8	23	46	77	116	163	218	281	352	431	518	613
566	475	392	317	250	191	140	97	62	35	16	5	6	7	22	45	76	115	162	217	280	351	430	517	612
567	476	393	318	251	192	141	98	63	36	17	18	19	20	21	44	75	114	161	216	279	350	429	516	611
568	477	394	319	252	193	142	99	64	37	38	39	40	41	42	43	74	113	160	215	278	349	428	515	610
569	478	395	320	253	194	143	100	65	66	67	68	69	70	71	72	73	112	159	214	277	348	427	514	609
570	479	396	321	254	195	144	101	102	103	104	105	106	107	108	109	110	111	158	213	276	347	426	513	608
571	480	397	322	255	196	145	146	147	148	149	150	151	152	153	154	155	156	157	212	275	346	425	512	607
572	481	398	323	256	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	274	345	424	511	606
573	482	399	324	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	344	423	510	605
574	483	400	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	422	509	604
575	484	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	508	603
576	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	602
577	578	579	580	581	582	583	584	585	586	587	588	589	590	591	592	593	594	595	596	597	598	599	600	601

12. Unique Prime Years

Program Name: Unique.java Input File: unique.dat

Judges Input File

10
2053
2012
2087
2099
2001
2081
2011
2069
2017
2027

Judges Output to Screen

2053 PRIME YEAR AND THE SUM OF 9 CONSECUTIVE PRIMES
2012 NOT A PRIME YEAR
2087 PRIME YEAR
2099 PRIME YEAR AND THE SUM OF MORE THAN ONE SET OF CONSECUTIVE PRIMES
2001 NOT A PRIME YEAR
2081 PRIME YEAR AND THE SUM OF 5 CONSECUTIVE PRIMES
2011 PRIME YEAR AND THE SUM OF MORE THAN ONE SET OF CONSECUTIVE PRIMES
2069 PRIME YEAR AND THE SUM OF 27 CONSECUTIVE PRIMES
2017 PRIME YEAR
2027 PRIME YEAR AND THE SUM OF MORE THAN ONE SET OF CONSECUTIVE PRIMES

Note to judges: these are the prime years and unique prime years in the 21st century

2003 prime

$157 + 163 + 167 + 173 + 179 + 181 + 191 + 193 + 197 + 199 + 211 = \mathbf{2011}$

$661 + 673 + 677 = \mathbf{2011}$

2017 prime

$29 + 31 + 37 + 41 + 43 + 47 + 53 + 59 + 61 + 67 + 71 + 73 + 79 + 83 + 89 + 97 + 101 + 103 + 107 + 109 + 113 + 127 + 131 + 137 + 139 = \mathbf{2027}$

$101 + 103 + 107 + 109 + 113 + 127 + 131 + 137 + 139 + 149 + 151 + 157 + 163 + 167 + 173 = \mathbf{2027}$

2029 prime

2039 prime

$199 + 211 + 223 + 227 + 229 + 233 + 239 + 241 + 251 = \mathbf{2053}$

2063 prime

$19 + 23 + 29 + 31 + 37 + 41 + 43 + 47 + 53 + 59 + 61 + 67 + 71 + 73 + 79 + 83 + 89 + 97 + 101 + 103 + 107 + 109 + 113 + 127 + 131 + 137 + 139 = \mathbf{2069}$

$401 + 409 + 419 + 421 + 431 = \mathbf{2081}$

$67 + 71 + 73 + 79 + 83 + 89 + 97 + 101 + 103 + 107 + 109 + 113 + 127 + 131 + 137 + 139 + 149 + 151 + 157 = \mathbf{2083}$

2087 prime

2089 prime

$13 + 17 + 19 + 23 + 29 + 31 + 37 + 41 + 43 + 47 + 53 + 59 + 61 + 67 + 71 + 73 + 79 + 83 + 89 + 97 + 101 + 103 + 107 + 109 + 113 + 127 + 131 + 137 + 139 =$

2099

$53 + 59 + 61 + 67 + 71 + 73 + 79 + 83 + 89 + 97 + 101 + 103 + 107 + 109 + 113 + 127 + 131 + 137 + 139 + 149 + 151 = \mathbf{2099}$

